

- A. Please describe the policies adopted in a PRAM model when a CW is present. (1)

- B. What can be said about the time complexity of a PRAM-based application when the PRAM model has a bounded number of shared memory cells? (1)

- C. Could you please define Speedup according to Gustafson's law? What assumptions are associated with this law?" (1)

- D. Please discuss the Amdahl and Gustafson laws in terms of "scaling." (1)

- A. Coherent execution is not necessary for efficient parallelization. True/False? When? (1)

- B. **Three different forms of parallel execution: Superscalar, SIMD, and Multicore.**
Please describe the differences between these three forms of execution. (2)

- C. Please describe the tradeoffs between the silicon area dedicated to the cache and the processor contexts. (1)

A. Please describe what a stall is and how its effect can be reduced. (2)

B. Please describe what a performant parallel program will do to overcome the bandwidth limits. (2)

- A. What happens if a main() function that launched multiple Pthreads threads returns after launching pthread_exit()? (1)

- B. Can a programmer request more OpenMP threads than there are processors in a system? How is the program executed in that case? (1) **This question can be skipped if you got 1 or 3 points in the first challenge**

- C. Describe the effects of the following pragma. (2) **This question can be skipped if you got 2 or 3 points in the first challenge.**

```
#pragma omp task depend(in: status[i])
```
